THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rhombus has side 25 cm and one diagonal 30 cm. Its area is: 1

(a) 750 cm² **(b)** 600 cm² **(c)** 300 cm² **(d)** 100 cm²

2. A path 2 m wide runs around the outside of a 40 m by 25 m rectangular garden. The area of the path is: 1

(a) 276 m² **(b)** 16 m² **(c)** 1276 m² **(d)** 260 m²

3. A right triangle has hypotenuse 40 cm and one leg 24 cm. Its area is: 1

(a) 640 cm² **(b)** 480 cm² **(c)** 384 cm² **(d)** 768 cm²

4. The diagonal of a square is 20 cm. Its area is: 1

(a) 40 cm² **(b)** 100 cm² **(c)** 200 cm² **(d)** 400 cm²

5. An L-shaped figure is made by removing a 14 cm by 8 cm rectangle from a 28 cm by 22 cm rectangle. Its area is: 1

(a) 728 cm² **(b)** 504 cm² **(c)** 196 cm² **(d)** 616 cm²

6. A parallelogram has base 28 cm and height 15 cm. If the base is changed to 21 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 17 **(b)** 399 **(c)** 20 **(d)** 23

7. A rhombus has side 20 cm and one diagonal 24 cm. Its area is: 1

(a) 80 cm² **(b)** 384 cm² **(c)** 480 cm² **(d)** 192 cm²

8. A path 3 m wide runs around the outside of a 35 m by 20 m rectangular garden. The area of the path is: 1

(a) 366 m² **(b)** 36 m² **(c)** 330 m² **(d)** 1066 m²

9. A right triangle has hypotenuse 53 cm and one leg 28 cm. Its area is: 1

(a) 742 cm² **(b)** 630 cm² **(c)** 1192 cm² **(d)** 1260 cm²

10. The diagonal of a square is 24 cm. Its area is: 1

(a) 144 cm² **(b)** 576 cm² **(c)** 288 cm² **(d)** 48 cm²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A square garden has a diagonal of 26 m. Find its area and side length (leave the side as a surd). 2

12. A rectangle has perimeter 120 cm and length 35 cm. Find its breadth and area. 2

13. A rhombus has side 35 cm and one diagonal 42 cm. Find its other diagonal and area. 2

14. A square garden has a diagonal of 30 m. Find its area and side length (leave the side as a surd). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular park is 35 m by 25 m. A path 1 m wide runs inside along its boundary. Find the area of the path. 3

16. A parallelogram has base 45 cm and height 8 cm. Another side of 30 cm has a corresponding height; find that height and the area. 3

17. A right triangle has legs 24 cm and 7 cm. Find its hypotenuse and area, and the length of the altitude drawn to the hypotenuse. 3

OR

A rectangular park is 48 m by 32 m. A path 2 m wide runs inside along its boundary. Find the area of the path. 3

18. A parallelogram has base 42 cm and height 10 cm. Another side of 35 cm has a corresponding height; find that height and the area. 3

OR

A right triangle has legs 16 cm and 30 cm. Find its hypotenuse and area, and the length of the altitude drawn to the hypotenuse. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A figure is made of a rectangle 40 cm by 15 cm with a right triangle of base 40 cm and height 20 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

OR

The parallel sides of a trapezium are 26 cm and 12 cm and its area is 266 cm². Find the distance between the parallel sides. A parallelogram of base 12 cm (the shorter parallel side) and the same height is cut out, leaving a triangle; find the area of the parallelogram and of the triangle left. 5

20. A figure is made of a rectangle 28 cm by 14 cm with a right triangle of base 28 cm and height 10 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

OR

The parallel sides of a trapezium are 30 cm and 22 cm and its area is 416 cm². Find the distance between the parallel sides. A parallelogram of base 22 cm (the shorter parallel side) and the same height is cut out, leaving a triangle; find the area of the parallelogram and of the triangle left. 5

END OF QUESTION PAPER — Total: 40 Marks